

Droplet Wall Interaction Tool: A Python Platform for Droplet Experiments and Qualitative Image Analysis

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Summary

Droplet hydrodynamics play a central role in many chemical engineering and industrial separation processes, including liquid–liquid extraction, multiphase reactors, and microfluidic applications. In such systems, the motion, deformation, and interaction of droplets with surrounding geometries strongly influence residence times, mass transfer, and overall apparatus performance. Droplets may accelerate, oscillate in shape, or experience significant changes in trajectory depending on whether they rise freely or interact with walls and internal structures such as packings or inserts. Experimental investigations of these effects require large numbers of high-speed video recordings under varying conditions. As a result, researchers are often confronted with extensive datasets consisting of hundreds of videos that must be evaluated consistently. Manual analysis is time-consuming and limits reproducibility. The Droplet Wall Interaction Tool was developed to address this challenge by providing an automated and extensible platform for droplet motion analysis across different experimental scenarios. The software enables quantitative characterization of droplet velocity, deformation, equivalent diameter, and contact line behavior. By integrating robust computer vision methods with an interactive graphical interface, the tool supports batch processing of large datasets while maintaining transparency and traceability of the analysis steps.

Statement of need

Understanding how wall contact and apparatus internals influence droplet motion requires systematic comparison against a well-defined baseline case. In particular, the freely rising droplet without geometric constraints represents an essential reference scenario. Only by quantifying this baseline behavior can researchers determine how droplets are altered by walls, channels, or structured packings. In experimental multiphase flow research, analysis workflows are often based on manual frame-by-frame evaluation or custom scripts developed for individual studies. Such approaches become impractical when datasets scale to hundreds of experiments and make it difficult to ensure consistent parameter selection and reproducible results. The Droplet Wall Interaction Tool was designed to provide a unified workflow for automated droplet characterization across multiple scenarios, including free droplet rise, droplet motion near inclined walls, confinement in channels, and interactions with structured surfaces. The software is intended for researchers in chemical engineering, surface science, multiphase transport, and microfluidics who require reproducible evaluation of droplet dynamics. By enabling automated batch analysis and standardized export formats, the tool supports systematic studies that would otherwise be limited by manual effort.

40 State of the Field

41 A wide range of algorithms and open-source packages exist for droplet contour detection,
42 image segmentation, and contact angle measurement. In particular, tools for sessile droplet
43 analysis on horizontal substrates are commonly used in surface science and wetting research.
44 These approaches are well suited for static and symmetric droplets under controlled laboratory
45 conditions. However, existing tools often become insufficient when droplets are moving,
46 deforming, or interacting with inclined or structured surfaces. Many available contact angle
47 packages rely on symmetry assumptions and are not designed for dynamically evolving droplet
48 shapes. Furthermore, most software solutions focus on isolated tasks such as static contact
49 angle determination and do not provide integrated analysis of droplet velocity, deformation
50 metrics, and contact line dynamics within a single framework. Another key limitation is
51 scalability. While some tools allow automated processing of individual videos, they typically
52 lack support for dataset-level batch evaluation, which is essential in experimental campaigns
53 producing hundreds of recordings. Allan et al. (2025) The Droplet Wall Interaction Tool
54 addresses these gaps by combining droplet motion tracking, deformation characterization, and
55 contact line analysis in a unified workflow. Its ability to analyze asymmetric, moving droplets
56 on inclined surfaces represents a distinct scholarly contribution beyond existing alternatives.
57 By integrating baseline free-rise experiments with wall-interaction modes in the same interface,
58 the tool enables systematic comparisons that are difficult to achieve with currently available
59 packages.

60 Software design

61 The Droplet Wall Interaction Tool was developed with the goal of providing an accessible
62 and reproducible workflow for experimental droplet hydrodynamics research. A central design
63 requirement was to combine robust computer vision algorithms with an interface that allows
64 researchers to process large datasets efficiently while maintaining transparency of the underlying
65 analysis.

66 The software is implemented in Python, ensuring that it can be easily executed, inspected,
67 and adapted by scientists without reliance on proprietary environments. This choice supports
68 extensibility and encourages reuse in related experimental studies.

69 Droplet detection and contour extraction are based on established methods from the OpenCV
70 ecosystem. In particular, edge-based segmentation using the Canny algorithm provides a reliable
71 foundation for identifying droplet boundaries under varying imaging conditions. Building on
72 these well-tested techniques enables robust droplet recognition across experimental scenarios,
73 including freely rising droplets and droplets interacting with inclined or structured surfaces.

74 A key contribution of the software is the integration of these algorithms into an interactive
75 graphical user interface (GUI). Many existing droplet analysis workflows rely on scripts that
76 are difficult to scale or reproduce when hundreds of experiments must be evaluated. The GUI
77 was therefore designed not only for usability, but also to enable dataset-level batch processing.
78 Users can define regions of interest, adjust baseline parameters, and immediately verify how
79 selected settings influence droplet detection.

80 Importantly, the interface provides real-time visual overlays of detected contours and
81 intermediate processing steps. This allows researchers to confirm that droplets are
82 correctly recognized throughout the analysis and reduces the risk of unnoticed parameter
83 misconfiguration.

84 The image analysis workflow begins with preprocessing steps such as cropping, rotation, and
85 background correction to ensure consistent droplet visibility. Droplet contours are then extracted
86 and geometric features such as area and equivalent diameter are computed. Depending on the
87 experimental mode, baseline intersections are evaluated to determine contact line properties

and advancing or receding contact angles. Droplet center positions are tracked over time, enabling velocity calculations from trajectory data. Results are exported in standardized Excel formats, while overlay images are saved to provide qualitative traceability and documentation of the automated evaluation.

Research impact statement

The Droplet Wall Interaction Tool has already been successfully applied in research on liquid–liquid extraction and droplet–wall interaction phenomena.

Mielke et al. used the software to quantify the influence of stainless-steel wall contact on the velocity of butyl acetate droplets. Their results showed that droplet deformation oscillations occur during free rise and that these oscillations are significantly reduced when droplets interact with a wall surface. (Mielke et al., 2025)

Pötter et al. applied the tool to investigate the invasiveness of film thickness sensors in confined channel experiments. By comparing droplet velocities of different organic phases, including butanol, silicone oil, and butyl acetate, on both stainless-steel and polymer sensor surfaces, they demonstrated that the sensor material does not significantly alter droplet hydrodynamics. This supports the applicability of such sensors for monitoring industrial apparatuses without disturbing flow behavior. (Pötter et al., 2026)

These applications demonstrate that the software already contributes to reproducible experimental workflows and provides a foundation for ongoing studies in droplet dynamics and multiphase transport.

AI usage disclosure

Generative AI tools were used during the development of this project. AI assistance was applied to improve code documentation by adding explanatory comments and to support language refinement of the manuscript, as English is not the native language of the authors.

The authors confirm that all AI-assisted outputs were carefully reviewed, edited, and validated by the human authors. All scientific interpretations, software design decisions, and final text content remain the responsibility of the authors.

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